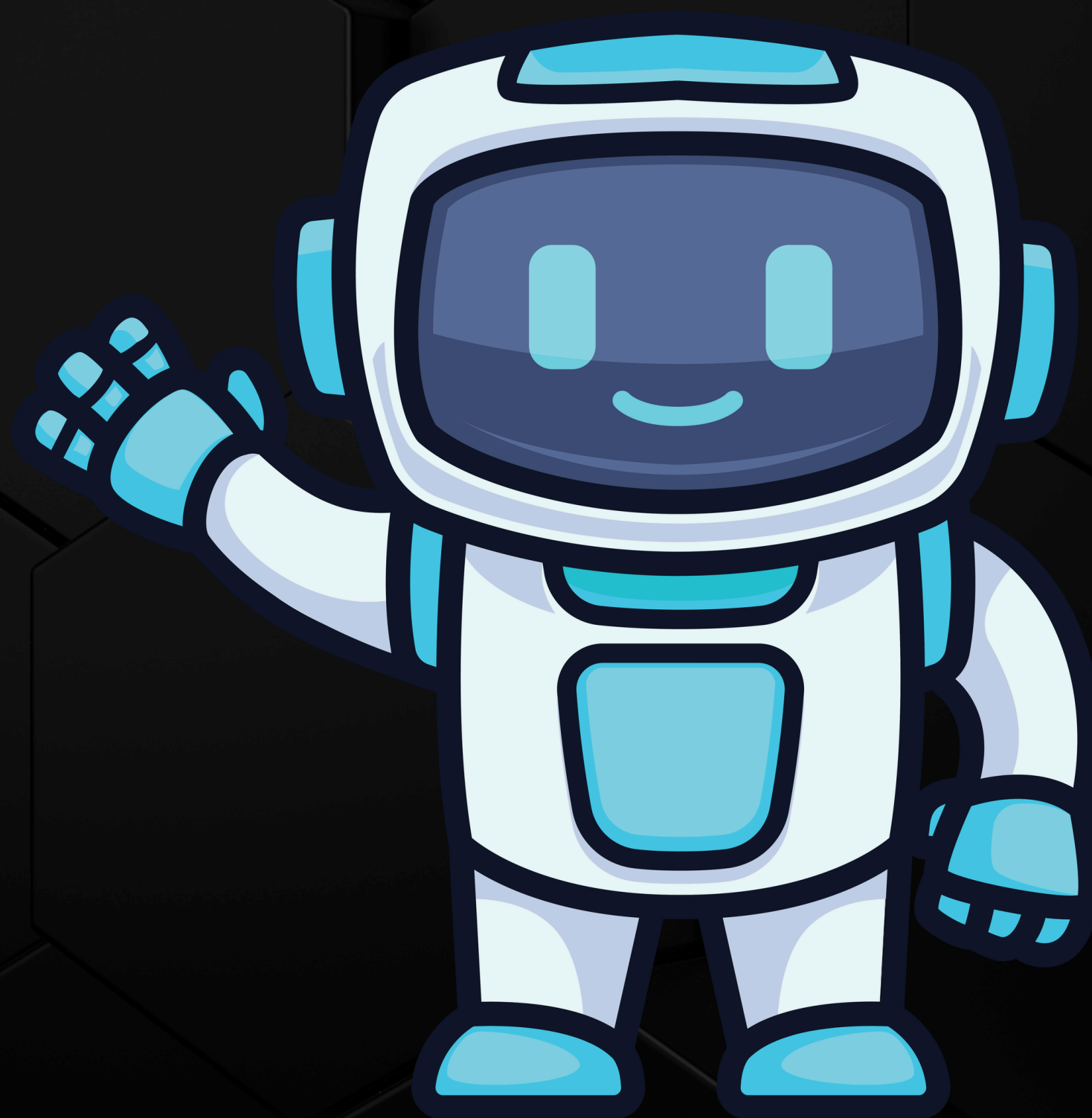




SMOKING CHECKER

by XGboost team

WWW.REALLYGREATSITE.COM





INSPECT AND PREPARE



Inspection: Gamma gt is right skewed

fix: by using loglp function

inspection: POC GT inaccurate correlation with Gamma gt levels

fix: Correcting rapid-test Gamma-GT by cartridgebatch :

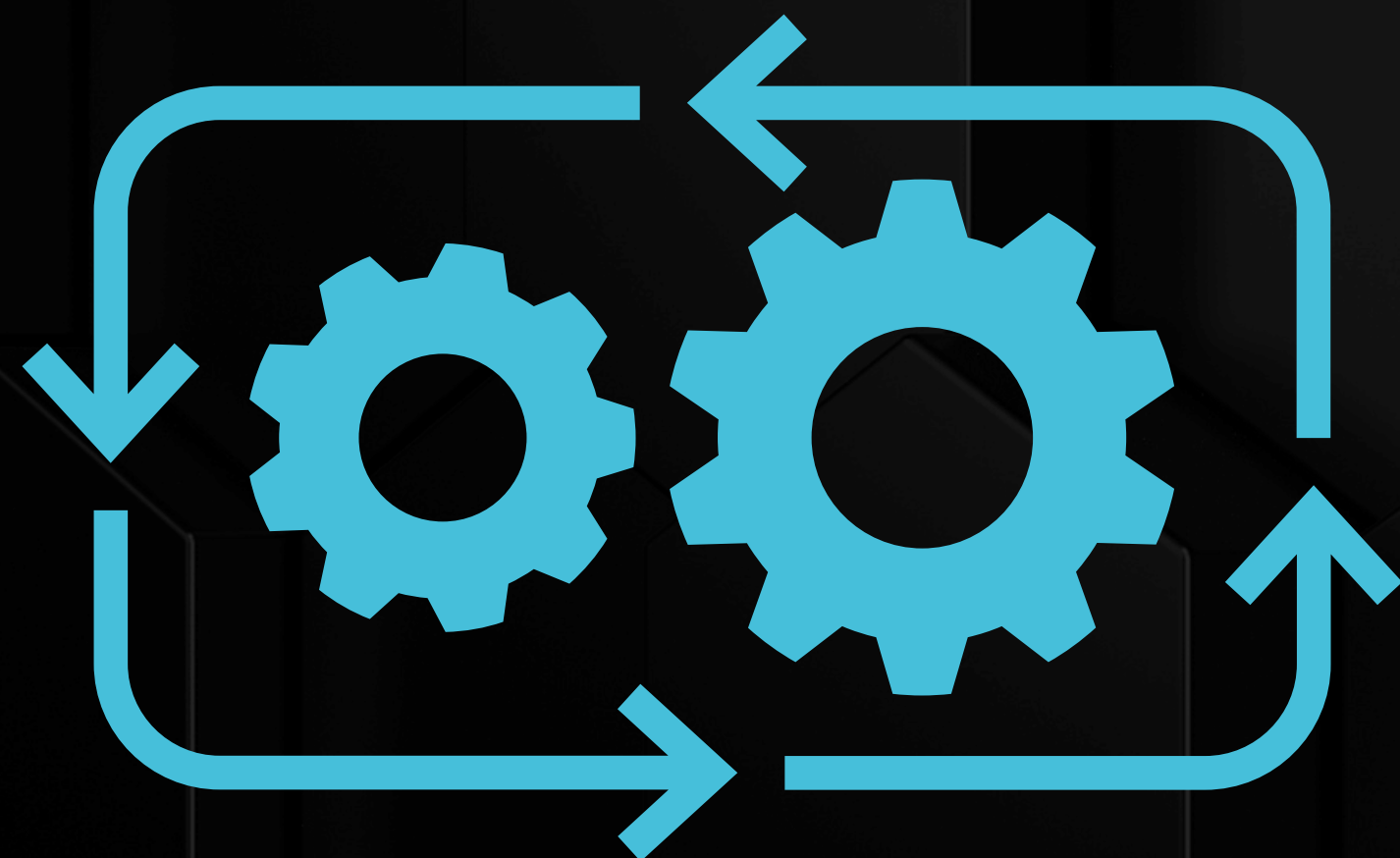
$\log(\text{Gamma_Gt}) = b * \log(\text{POC_GGT}) + \text{batch offset}$

inspection: missing CO data

fix: flagging the centers where CO is missing

inspection: Biomarkers inconsistent measurements

fix: Used urine creatinine to normalize relevant urine measurements, including cotinine and EtG.



VALIDATION

- 1-CV DESIGN
- 2-out-of-fold feature generation
- 3-local validation
- 4-final model selection

MODEL DEVELOPMENT

The combined CV score improved from 0.72899 with baseline CatBoost models to 0.81332 after batch calibration. Further feature engineering, detection-floor refinements, multi-seed training, neural-network blending, and precision calibration steadily increased the score to 0.86116. These results suggest that measurement corrections and model ensembling improved performance, though the experiments were not all directly comparable.



THYNK
UNLIMITED

COMMUNICATION AND TEAM WORK





THYNK
UNLIMITED



FUTURE DIRECTIONS

WWW.REALLYGREATSITE.COM